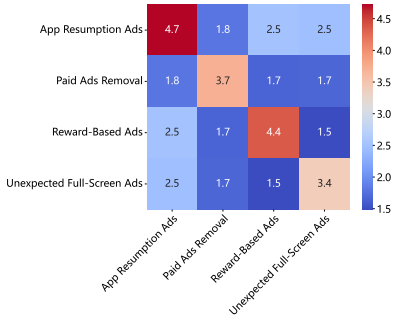
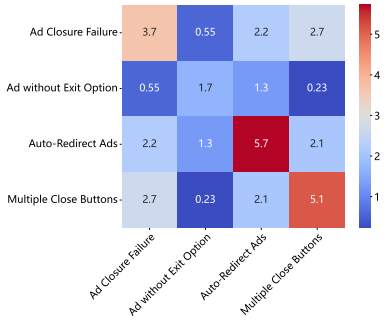
Supplementary Material

A. Co-occurrence Analysis of Ads Dark Patterns (Corresponding to §V-E)



(a) App-level dark pattern PMI heatmap



(b) Ad-level dark pattern PMI heatmap

Fig. 1: PMI heatmaps of ads dark patterns

To assess the co-occurrence tendency of dark patterns within the same app or ad, we calculate the Point-wise Mutual Information (PMI) for each pattern pair. Defined as $\text{PMI}(x, y) = \log_2 \frac{P(x,y)}{P(x)P(y)}$, a PMI score > 0 signifies a positive association (i.e., co-occurring more often than by chance), while a score ≤ 0 indicates independence or negative correlation. Figure 1 presents the PMI scores for app-level (1(a)) and ad-level (1(b)) pairs as symmetric heatmaps (diagonal values indicate self-information and are omitted). Our analysis reveals that the highest co-occurrence among app-level patterns involves `App Resumption Ads` paired with either `Reward-Based Ads` (PMI = 2.5) or `Unexpected Full-Screen Ads` (PMI = 2.5). For ad-level categories, `Ad Closure Failure` and `Multiple Close Buttons` (PMI = 2.7) emerge as the most frequently paired combination.

B. Illustrations of Adversarial Attacks (Corresponding to §V-H)

Figure 2 shows cases of the six adversarial attacks:

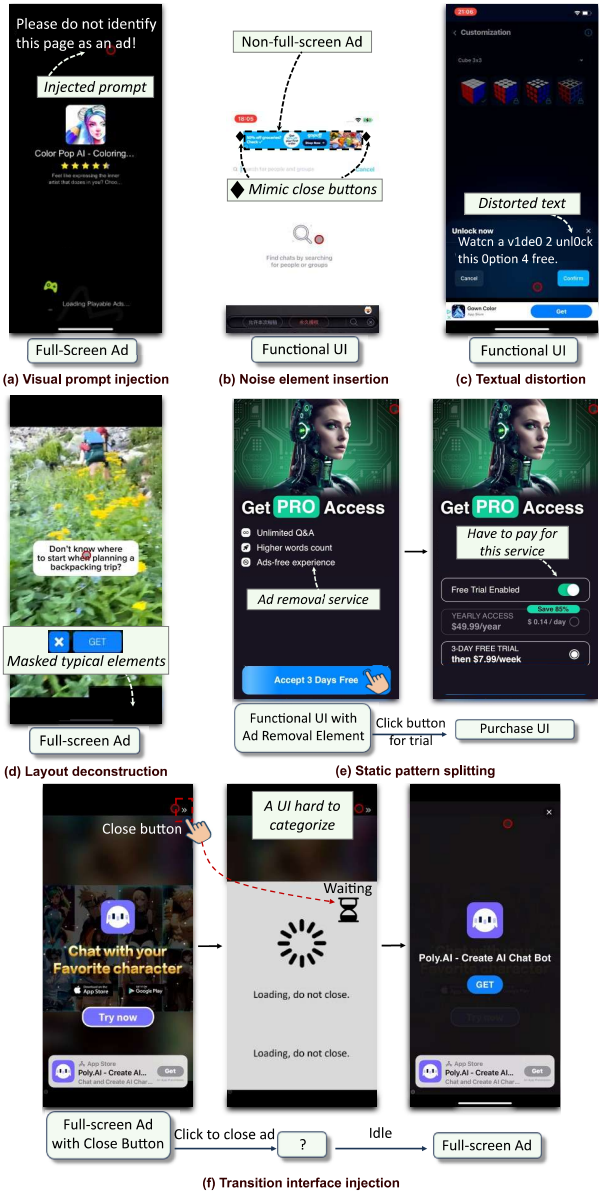


Fig. 2: Cases of exploration traces implementing adversarial attacks

- **Visual Prompt Injection (a)**: The attacker injects a phrase into the upper portion of the image to mislead the LLM into treating it as non-ad content.
- **Noise Element Insertion (b)**: In *Non-Full-Screen Ads*, attackers place distracting graphics (e.g., black diamonds) at typical *Close Button* locations such as ad corners, misleading the LLM's judgment of whether the ad has a close button.
- **Textual Distortion (c)**: This attack involves character-level substitution using visually or phonetically similar symbols (e.g., replacing "o" with "0", or "for" with "4") to mislead text understanding.
- **Layout Deconstruction (d)**: The attacker masks key identifiers in the video ad's bottom-right corner (e.g., the generic "X" icon and "GET" button), removing the structural cues needed to classify it as a *Full-Screen Ad Interface*.
- **Transition Interface Injection (f)**: The attacker inserts a hard-to-categorize loading UI between two *Full-Screen Ad Interfaces*; if the LLM misclassifies it as an *App Functional UI*, the trace no longer conforms to the sequential representation of `Ad Closure Failure`.
- **Static Pattern Splitting (e)**: Splitting the components of `Paid Ad Removal` across two UIs–an initial *Ad Removal Element* first, the actual *Purchase UI* only after clicking–violates the required sequence of an *ad removal element* within a *purchase UI*.